EECE 2311

Lab for EECE2310: Introduction to Digital Design and Computer Architecture

SPRING 2026

Lab Assignment #1

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EECE2311 Lab Section #

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Abstract

This lab is designing and verifying binary adder circuits using Quartz's Prime Schematic tool and DE1-SoC FPGA board. A half adder and a full adder were constructed using circuit design based on logic gates.

Introduction

The objective of this lab was to become familiar with the Quartus Prime environment and the hardware features of the Cyclone V FPGA on the DE1-SoC board. Specifically, the lab focused on building and testing a Half Adder—a circuit capable of adding two 1-bit numbers—and a Full Adder—which adds three 1-bit numbers.

Lab Setup

Pre-Lab

The Half Adder logic was analyzed to determine the Sum (S) and Carry out (Cout) for all input combinations.

A	*B*	*S*	C_{out}
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Equipment

Software: Quartus Prime Lite Edition.

Hardware: DE1-SoC Board (Cyclone V FPGA 5CSEMA5F31C6).

Input/Output Devices: Slide switches (SW[0]–SW[2]) and LED lights (LEDR[0], LEDR[1]).

Connectivity: USB-A/USB-B cable and a power adapter

Results and Analysis

This section may be divided into "Results" and "Analysis" sections if there are many results to discuss. For multi-part labs, you should include subsections to discuss each part.

Results for half adder

Half_adder schematic

This half-adder schematic takes two inputs, sw[0] (A) and sw[1] (B), and generates two outputs: Sum and Carry. The inputs are first inverted using NOT gates to create A' and B', which are then combined with AND gates to form A'B and AB'. These two terms are ORed together to produce the Sum, implementing the XOR function. In parallel, another AND gate directly combines A and B to generate the Carry output, which is high only when both inputs are high.

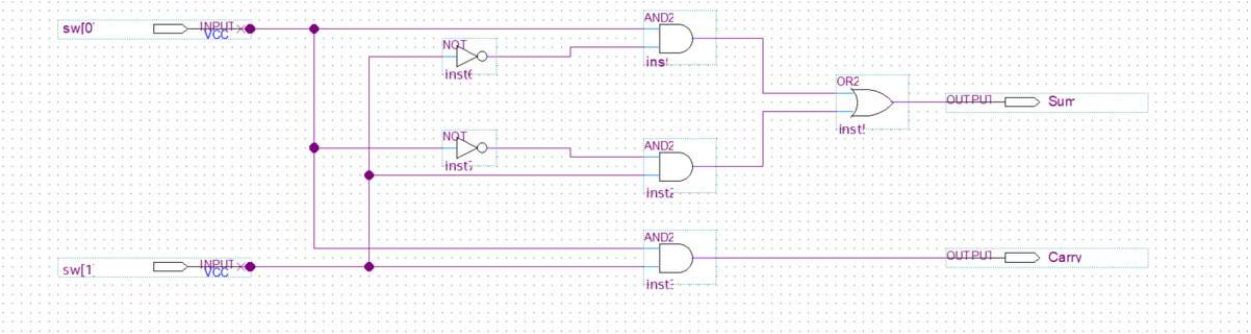


Fig. 1: Half_adder schematic

The waveform of the half-adder shows how the Sum and Carry outputs respond as the two inputs A (sw[0]) and B (sw[1]) toggle through all combinations. When both inputs are low (A=0, B=0), both Sum and Carry remain low. When either input goes high while the other stays low (A=1, B=0 or A=0, B=1), the Sum output goes high while the Carry stays low, reflecting the XOR behavior. When both inputs are high (A=1, B=1), the Sum returns to low and the Carry goes high, indicating a carry generation. Overall, the waveform shows Sum pulses whenever the

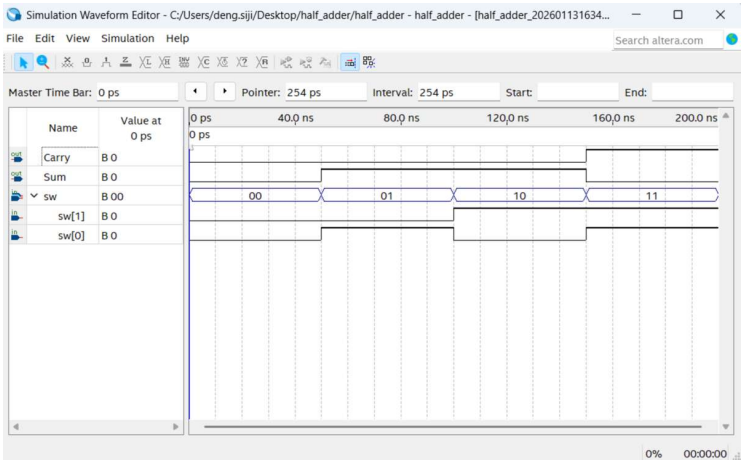


Fig. 2: Simulation results: half_adder waveform

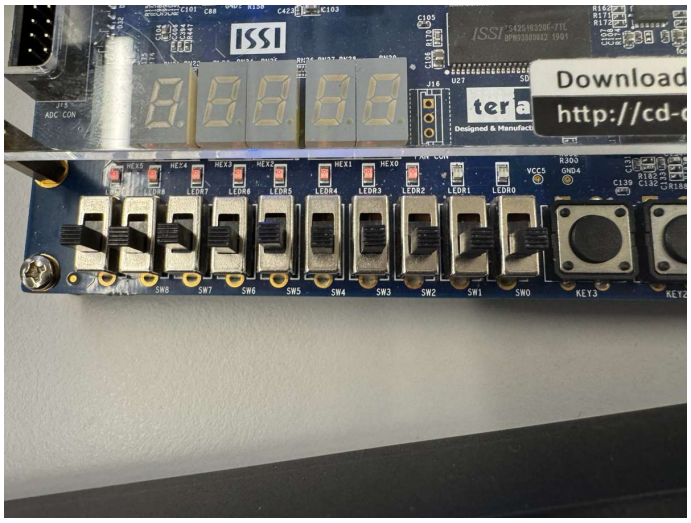


Fig 3. Input 0, 0; output sum 0, carry 0

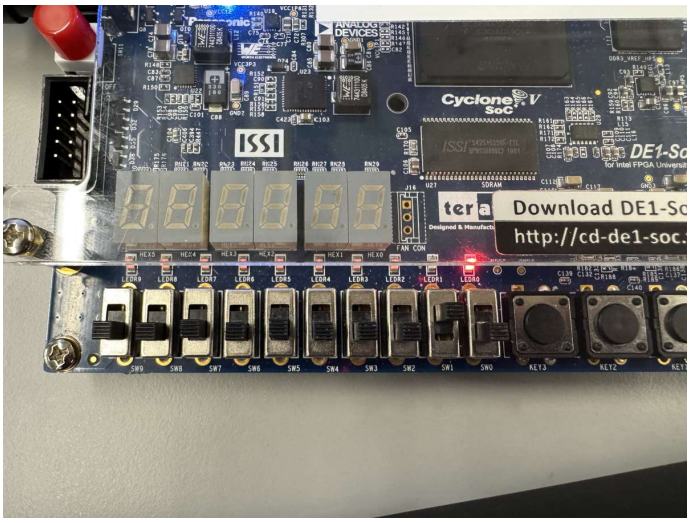


Fig 4. Input 1, 0; output sum 1, carry 0

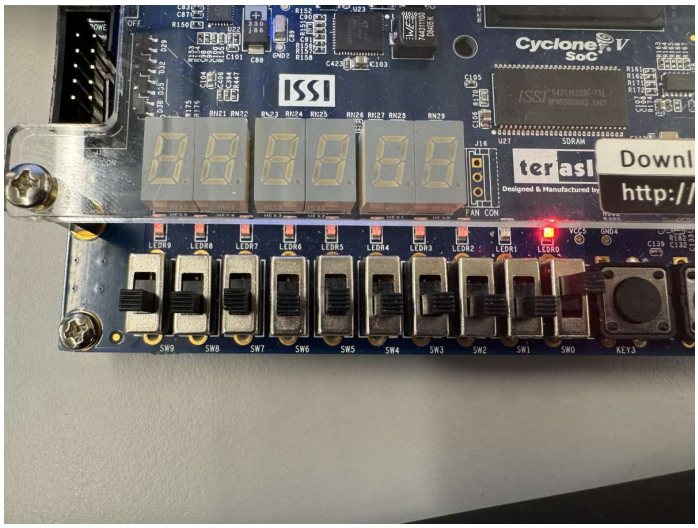


Fig 5. Input:0,1; output: sum 1, carry 1

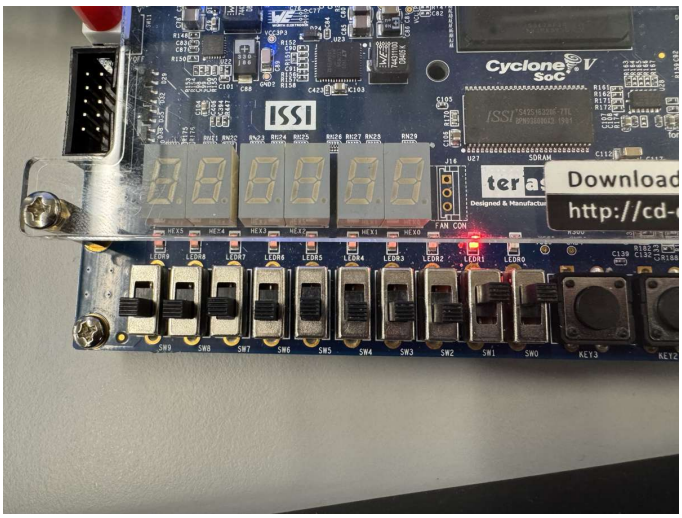


Fig 6. Input 1, 1; output sum 0 carry 1

sw[1]	sw[1]	Sum LEDR[0]	Carry LEDR[1]
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Table 1. 2-input 1-digit half adder result

Analysis

As shown in the table, the behavior of the board switches and indicator lights for a half-adder. When both switches sw[1] and sw[0] are OFF (0,0), neither the Sum nor the CarryOut light turns on, indicating no addition result. When exactly one switch is ON (0,1 or 1,0), the Sum light turns ON while the CarryOut remains OFF, showing that the half-adder produces a sum of 1 with no carry. When both switches are ON (1,1), the Sum light turns OFF and the CarryOut light turns ON, indicating that the sum exceeds 1 and a carry is generated.

The results validate the half-adder design and meet expectations. The circuits design is successful and correct.

The design correctly accounts for both the input bits and the carry-in, validating the design and implementation.

Results for full adder

Full_adder truth table

X	Y	Z	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Fig 7 full adder truth table

Full_adder schematic

This full-adder schematic takes three inputs, A, B, and Cin (carry-in), and produces two outputs: Sum and Carry-out. The circuit first combines A and B using an XOR operation to generate an intermediate sum, which is then XORed with Cin to produce the final Sum output. In parallel, carry logic is formed using AND gates to detect when pairs of inputs are high (A·B,

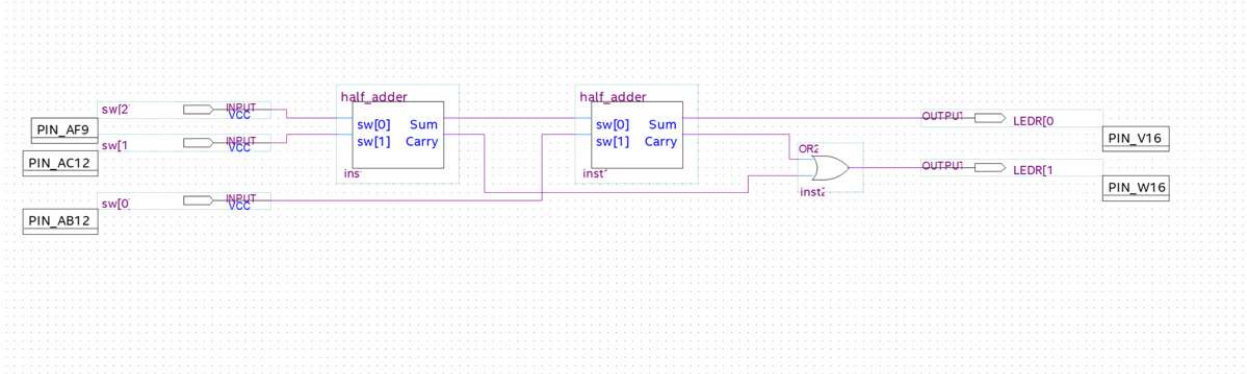


Fig. 8: Full_adder schematic

A·Cin, and B·Cin). These carry terms are then combined using OR gates to generate the Carry-out, which goes high whenever two or more of the inputs are high.

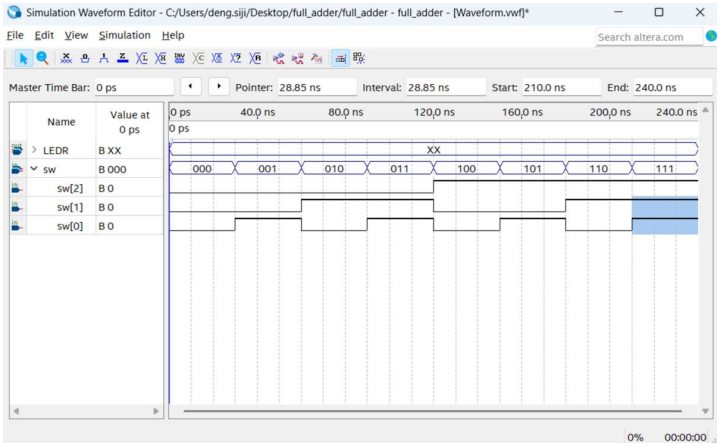


Fig 9: Simulation result: full adder waveform

A full adder can be built using two half adders and one OR gate. In this design, the first half adder adds the inputs A and B, producing an intermediate sum (S_1) and a carry (C_1). The second half adder then adds S_1 and the carry-in (Cin), generating the final Sum output and a second carry (C_2). Finally, the two carry outputs (C_1 and C_2) are combined using an OR gate to produce the Carry-out.

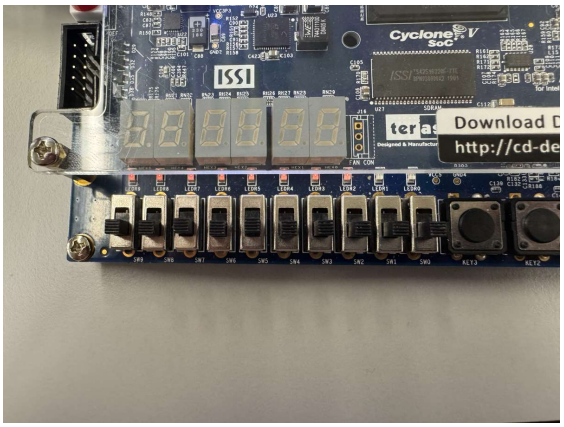


Fig 10. Input (0,0,0); output sum 0 carry 0

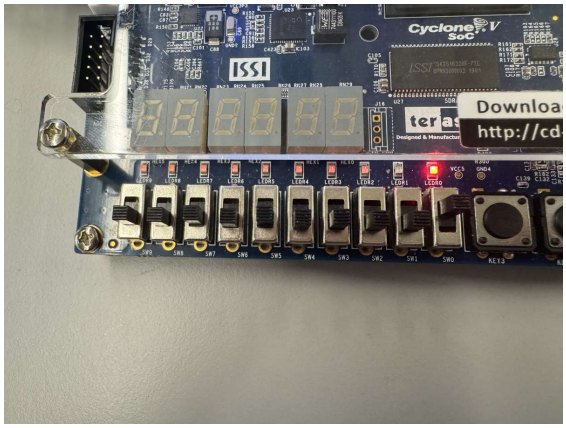


Fig 11. Input (0,0,1); Output sum 1 carry 0

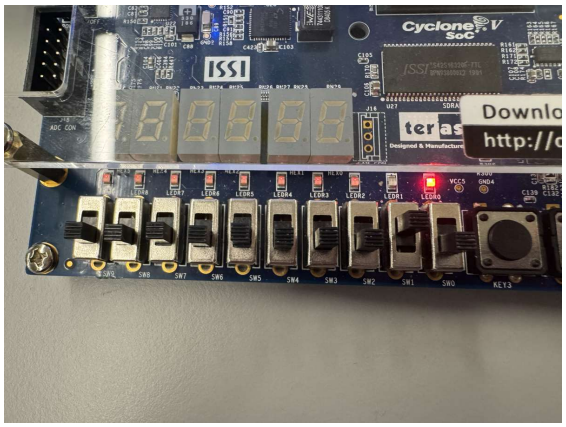


Fig 12. Input (0,1,0);
Output sum 1 carry 0

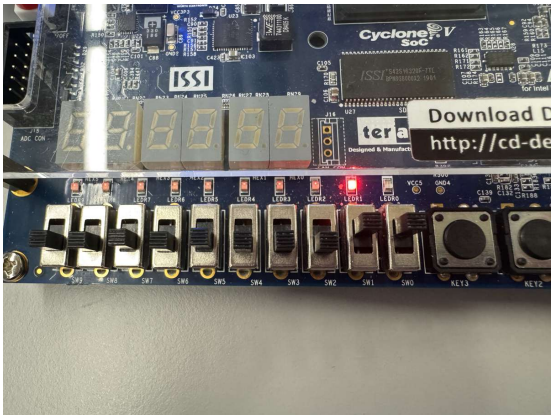


Fig 13. Input (0,1,1);
Output sum 0, carry 1

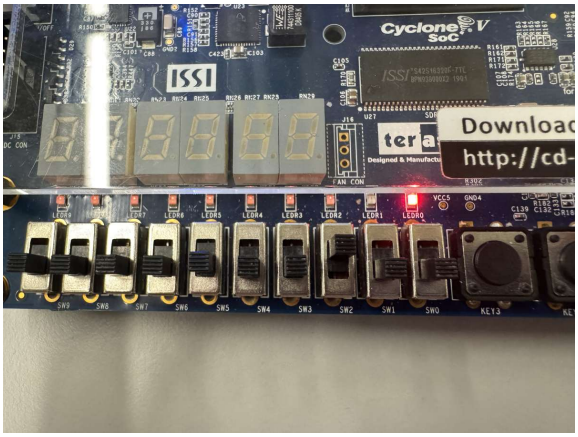


Fig 14. Input(1,0,0);
Output sum 1, carry 0

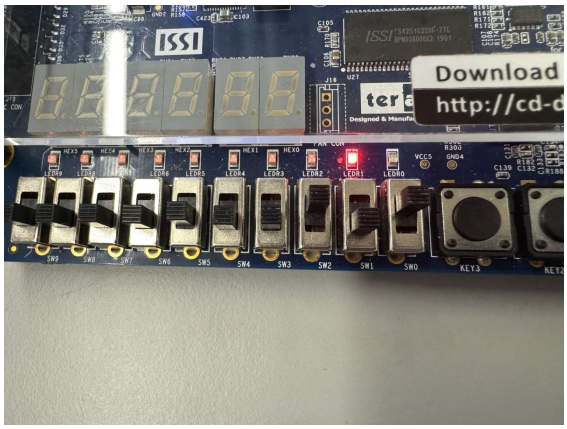


Fig 15. Input(1,0,1);
Output sum 0,carry 1

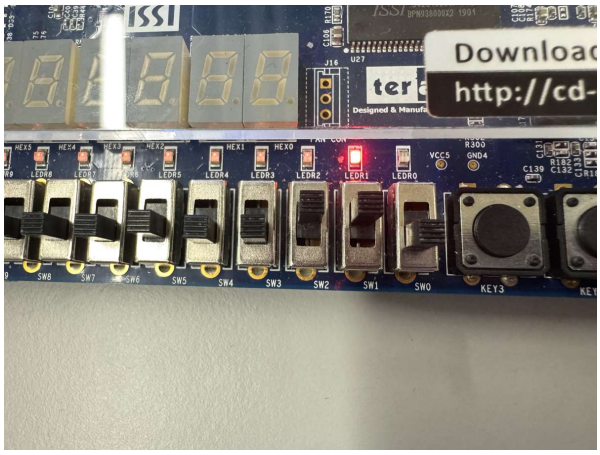


Fig 16. (1,1,0);
Output sum 0, carry 1

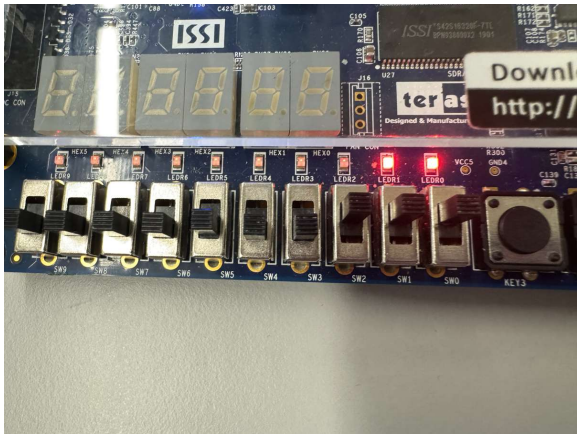


Fig 17. (1,1,1);
Output sum 1, carry 1

Simulation result

sw[2]	sw[2]	sw[2]	SumLEDR[0]	CarryLEDR[1]
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Table 2. 3-input 1digit full adder result

Analysis

The full adder results shown in the table and board images confirm correct operation of a 3-input, 1-bit full adder using switches sw[2], sw[1], and sw[0] as inputs. When all three inputs are 0, both the Sum and Carry outputs remain 0, indicating no addition. When exactly one input is high, the Sum output turns on while the Carry remains off, representing a sum of 1 with no carry. When two inputs are high, the Sum turns off and the Carry turns on, showing that the total exceeds 1 and generates a carry. Finally, when all three inputs are high (1,1,1), both Sum and Carry are high, representing a binary result of 3 (11_2).

The LED outputs on the board match the expected truth table.

The full adder correctly accounts for both the input bits and the carry-in, validating the design and implementation.

Conclusion

The design correctly accounts for both the input bits and the carry-in, validating the design and implementation.

The LED outputs on the board match the expected truth table.

The full adder correctly accounts for both the input bits and the carry-in, validating the design and implementation.

References

1. Michael Benjamin, "*Lab Report Guide*", Northeastern University, January 18, 2006.
2. Given lab 1 pdf in EECE2311 canvas.